

Artificial Intelligence in Transportation: A Cross-Domain Review of Predictive Maintenance

Ismayil Hasanli A, Leyla Hasanova B

Abstract:

The role of artificial intelligence (AI) in enhancing predictive maintenance (PdM) becomes increasingly evident among the key segments of transportation, such as aviation, marine transportation, and automotive production. Employing various machine learning techniques like k-Nearest Neighbors (kNN), LSTM networks, Support Vector Machines (SVM), Random Forests, and autoencoders enables PdM based on AI to detect faults, predict the Remaining Useful Life (RUL), and extract valuable information for maintenance purposes from continuous streams of data obtained from sensors. The paper presents a structured overview of the implementation of AI-supported PdM in three areas of transportation by summarizing recent empirical and review articles on the topic (five articles). The mathematical aspects related to classification techniques and performance of AI-powered PdM, i.e., the confusion matrix, classification accuracy measure, Risk Priority Number (RPN), and trend analysis, are presented in conjunction with results achieved. Empirical findings show that the use of AI-driven predictive maintenance decreases unexpected downtime by 30%, decreases expenses associated with maintenance operations by about 25%, and provides classification accuracy rates above 90%.

Keywords: predictive maintenance, artificial intelligence, machine learning, kNN, LSTM, maritime, aviation, automotive, fault diagnosis, remaining useful life

1. Introduction

The transport infrastructure itself forms the foundation of the global economy, and any unplanned downtime in equipment used either at sea, air, or on roads can cause huge safety and economic impacts. The industries of Aviation is estimated to waste over 20% of its maintenance expenditure on unscheduled maintenance, and 5% more on fuel consumption due to lack of maintenance of its

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machinery (Patibandla, 2024). While it has been estimated that there were over 11,500 maritime incidents recorded between 2014 and 2023 caused by equipment failure (four times more than the number of collisions), and equipment failure accounted for the leading cause of maritime incidents recorded between 2014-2023 (more than 50% of maritime incidents in 2023 were due to equipment failure). As compared with automotive industries, the high level of automation in automotive production line makes even smallest downtime of machinery impact production steps of the supply chain, which is considered as the most critical risk (Andersson et al., 2025).

The traditional concepts of corrective and preventive maintenance that have dominated transportation industries have proven ineffective to meet stringent demands in safety and performance. Corrective maintenance requires inspection and repair or replacement once a failure occurred, while preventive maintenance requires the maintenance of components after a predetermined interval irrespective of actual component condition, thus leading to over-maintainability and miss pre-failure symptoms (Patibandla, 2024; Johari et al., 2026). The predictive maintenance (PdM) paradigm which is used to model future behavior of machinery and predict an imminent failure event using the data from real time sensors and intelligent algorithms has overcome the above problems; maintenance can then be performed at the precise moment predicted by algorithms to have a forthcoming equipment failure, which thereby minimizes unpredicted maintenance interventions and emergencies.

The application of machine learning (ML) algorithms has provided a great advancement toward the functionality of PdM systems. Different from the physics-based models where engineer's profound knowledge of failure mechanisms is necessary, ML models are capable of learning relationships between operating sensor measurements and machinery health or component lifetime without explicitly defining certain physical relations or lack of knowledge about failures. ML models can be divided into supervised learning models such as kNN, SVM, Random Forest to predict failure categories; recurrent learning models such as LSTMs, widely applied to forecast degradation trends; and unsupervised learning models such as autoencoders, often used for anomaly detection with limited failure information, etc. The above machine learning models provide an appropriate paradigm to build up the flexible and powerful PdM toolbox for applications over different domains of transportation. (Pavlyuk & Alomar, 2026).

This article presents a detailed review of the AI-driven PdM techniques application in various transportation fields (Aviation, Maritime and Automotive) using 5 relevant, peer-reviewed articles published between 2024-2026. In section 2, methodology for selecting relevant literature is described. Mathematical fundamentals of some core ML algorithms and empirical results are presented in section 3, and potential research and industry applications are suggested in section 4.

2. Research Methods

2.1. Study Design and Source Selection

In this article, a structured narrative review approach is employed. Five main sources were chosen as a sample representative of the current status of AI-enabled PdM in the 3 transportation sectors addressed in the scope: (i) a preprint on the use of AI-driven PdM in marine transportation (Simion et al., 2024); (ii) a practitioner focused paper concerning AI based aviation maintenance at Southwest airlines (Patibandla, 2024); (iii) an experimental paper discussing a proposed MOMENT system for the diagnosis of automobile air filter using OBD-II data (Johari et al., 2026); (iv) a quantitative investigation using a combination of LSTM, SVM and autoencoder for application on a real industrial scenario of automotive production (Andersson et al., 2025); and (v) a systematic literature review of 696 aircraft maintenance related papers in the period 1984-2024 that are discussing the use of AI in this field (Pavlyuk & Alomar, 2026).

The selection criteria specified that each source (a) was published/posted in a peer reviewed or established academic journal/platform from 2024–2026, (b) was about at least one transportation sub-sector, (c) documented specific AI algorithms or model structures, and (d) produced either quantitative outcomes or a formal overview of empirical evidence. Together, the five sources represent experimental data, systems-level engineering proposals and macro-level bibliometric analysis.

2.2. Analytical Framework

Every source was characterized in four dimensions: (1) domain and scope of application; (2) applied AI technologies; (3) data used and pre-processed; and (4) performance measures reported. In order to allow a comparison across different domains, a general performance table was created and the mathematics behind main algorithms used were summarized using common notation. If empirical measures were reported in the source (accuracy, F1-score, downtime or cost savings), the given numbers were adopted; if it was a bibliometric or architectural study, qualitative synthesis were applied.

3. Findings and Discussion

3.1. Cross-Domain Overview

Table 1 summarizes the AI algorithms, data sources, and performance outcomes across the three transportation domains covered by the reviewed studies.

Domain	Study	Primary Algorithm(s)	Key Performance Outcomes
Maritime	Simion et al. (2024)	kNN, FMECA, FTA	12-class fault detection; real-time RUL monitoring

Domain	Study	Primary Algorithm(s)	Key Performance Outcomes
Aviation	Patibandla (2024)	SageMaker ML, IoT sensors	Reduced unplanned downtime; proactive scheduling
Aviation (SLR)	Pavlyuk & Alomar (2026)	LSTM, CNN, SVM, RF, Expert Systems	696 papers; DL dominant post-2018; LSTM & CNN lead
Automotive (air filter)	Johari et al. (2026)	kNN, SVM, Random Forest	kNN F1 = 0.983; 32,167-sample OBD-II dataset
Automotive (mfg.)	Andersson et al. (2025)	LSTM, SVM, RF, Autoencoder	92% accuracy; 30% less downtime; 25% cost savings

Table 1. Summary of AI-driven PdM across transportation domains

Although the operational environments of the three domains are different, the architectures across the three are very similar. They all use arrays of sensors that provide raw input to a cloud or edge computing layer, which is then followed by data pre-processing, feature extraction and model inference and results (alerts/maintenance advice) being provided to operators. This architecture can be clearly demonstrated with Patibandla's (2024) AWS-based aviation pipeline, Simion et al.'s (2024) five module maritime algorithm, and Andersson et al.'s (2025) automotive PdM pipeline.

3.2. Mathematical Foundations

3.2.1. Risk Priority Number (RPN) and FMECA

Failure Mode, Effects, and Criticality Analysis (FMECA) provides the structured failure taxonomy on which ML classifiers are trained. Its core quantitative measure, the Risk Priority Number (RPN), is defined as:

$$RPN = S \times O \times D \quad \dots (1)$$

where $S \in [1, 10]$ is the severity of the failure's effect on the system, $O \in [1, 10]$ is the probability of occurrence, and $D \in [1, 10]$ is the likelihood that the failure would go

undetected. The higher values of RPN correspond to failure modes that require more attention. Simion et al. (2024) developed a FMECA integrated with Fault Tree Analysis (FTA) for a seawater cooling system of an oil tanker and grouped 12 possible states-11 failure and 1 nominal-in a fault array used to train kNN. The traditional RPN has already drawbacks (non-unique products and equally important dimensions), so the weighted and fuzzy-logic based versions have been introduced in recent studies (Simion et al., 2024).

3.2.2. *k*-Nearest Neighbors (kNN) Classification

The kNN algorithm classifies a query point x by a majority vote among its k nearest training points under a chosen distance metric. Using the Euclidean distance in an n -dimensional feature space:

$$d(x_i, x_j) = \sqrt{(\sum_m (x_{im} - x_{jm})^2)} \quad \dots \quad (2)$$

the predicted class $\hat{y}$ for a new observation x is:

$$\hat{y} = \operatorname{argmax}_c \quad |\{x_j \in N_k(x) : y_j = c\}| \quad \dots \quad (3)$$

where $N_k(x)$ denotes the set of k nearest neighbors and c ranges over all classes. In the maritime application, numeric features were standardized (zero mean, unit variance) before fitting, and $k = 5$ was used (Simion et al., 2024). In MATLAB notation, the model is instantiated as:

$$Mdl = \text{fitcknn}(X, Y, 'NumNeighbors', 5, 'Standardize', 1) \quad \dots \quad (4)$$

The automotive MOMENT study (Johari et al., 2026) compared kNN against SVM and Random Forest on 28,226 OBD-II samples. With optimal hyperparameters found via GridSearchCV (5-fold cross-validation), validation F1-scores were: SVM = 0.925, RF = 0.979, kNN = 0.983, confirming kNN's superiority for this low-dimensional, well-separated classification problem.

3.2.3. Confusion Matrix and Classification Metrics

Model performance is evaluated through the confusion matrix. For a binary classifier, the matrix takes the form:

$$CM = [[TP, FP], [FN, TN]] \quad \dots \quad (5)$$

from which standard metrics are derived:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad \dots \quad (6)$$

$$Precision = TP / (TP + FP) \quad \dots \quad (7)$$

$$Recall = TP / (TP + FN) \quad \dots \quad (8)$$

$$F1\text{-score} = 2 \times (Precision \times Recall) / (Precision + Recall) \quad \dots \quad (9)$$

For multi-class problems (e.g., the 12-state maritime system), normalized confusion matrices allow comparison across classes of unequal size. Simion et al. (2024) compute the normalized confusion matrix as $CM_{\text{norm}} = \text{cm.NormalizedValues}$, expressing each cell as a proportion of its true class total. Andersson et al. (2025)

report aggregate accuracy of 92% for both SVM and Random Forest classifiers in the automotive manufacturing context, with matching precision and recall above 91%.

Model	Accuracy	Precision	Recall	F1-Score
SVM	92%	91%	93%	92%
Random Forest	92%	92%	91%	91.5%
kNN (MOMENT)	98.3%	—	—	0.983
LSTM (Automotive mfg.)	MAE = 0.12%	—	—	48-hr forecast
Autoencoder	90%	—	—	Anomaly detection

Table 2. Model performance metrics across reviewed studies

3.2.4. Long Short-Term Memory (LSTM) Networks and RUL Estimation

LSTM networks address the vanishing-gradient limitation of standard recurrent neural networks (RNNs) through gated memory cells. Each cell maintains a hidden state h_t and a cell state c_t updated by input, forget, and output gates (i, f, o). The cell dynamics at time step t are:

$$f_t = \sigma(Wf \cdot [h_{t-1}, x_t] + bf) \quad (\text{forget gate}) \quad \dots (10)$$

$$i_t = \sigma(Wi \cdot [h_{t-1}, x_t] + bi) \quad (\text{input gate}) \quad \dots (11)$$

$$c_t = \tanh(Wc \cdot [h_{t-1}, x_t] + bc) \quad (\text{candidate}) \quad \dots (12)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot c_t \quad (\text{cell state}) \quad \dots (13)$$

$$o_t = \sigma(Wo \cdot [h_{t-1}, x_t] + bo) \quad (\text{output gate}) \quad \dots (14)$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{hidden state}) \quad \dots (15)$$

where σ denotes the sigmoid function, $\odot$ is element-wise multiplication, and W and b are learnable weight matrices and bias vectors. Due to the long-term memory aspect of LSTMs they appear suitable for RUL prediction on time-series sensor data. Andersson et al. (2025) successfully applied a 12-month training period to sensor data from manufacturing automation equipment (temp, vibration, pressure, operational

velocity) and achieved an MAE of 0.12% for equipment failures up to 48 hours ahead of the failure, a predictive window which is not achieved by the SVM and Random Forest models (Pavlyuk & Alomar, 2026). LSTM is one of the two predominate architectures used in aircraft maintenance (Pavlyuk & Alomar, 2026), with there being 98 publications regarding this architecture up to 2024.

3.2.5. Linear Trend Modeling for Degradation Analysis

In the maritime algorithm, the ANALYSIS module fits a linear regression model to each sensor variable over time to estimate degradation trend direction and rate:

$$y_i(t) = \alpha_i + \beta_i \cdot t + \varepsilon_i(t) \quad \dots (16)$$

where y_i is the i -th sensor reading, α_i is the intercept, β_i is the slope (trend), and $\varepsilon_i(t)$ is zero-mean noise. The slope β_i is extracted via MATLAB's `fitlm` function:

$$\begin{aligned} mdl &= \text{fitlm}(\text{time}, \text{sensors}(:, i)) \\ \beta_i &= mdl.Coefficients.Estimate \quad \dots (17) \end{aligned}$$

A threshold τ is set for small variations. If $|\beta_i| < \tau$, the variable is classified as stable; positive slopes indicate increasing drift; negative slopes indicate decreasing drift. These trend labels serve as additional features for the kNN fault classifier and as early-warning signals for the LINK2PMS maintenance scheduling module (Simion et al., 2024).

3.2.6. Deviation Normalization in the DIAG Module

To create a unified, interpretable deviation metric across heterogeneous sensor dimensions, the DIAG module maps raw sensor values into the normalized range $[-100, +100]$ as follows:

$$\text{DiagArray}(i,j) = -100, \text{ if } \text{GraphData}(i,j) < \text{min_max}(1,j) \quad \dots (18a)$$

$$\text{DiagArray}(i,j) = (\text{GraphData}(i,j) - \text{min_max}(1,j)) / (\text{min_max}(2,j) - \text{min_max}(1,j)) \times (-100),$$

$$\text{if } \text{min_max}(1,j) \leq \text{GraphData}(i,j) < \text{min_max}(2,j) \quad \dots (18b)$$

$$\text{DiagArray}(i,j) = (\text{GraphData}(i,j) - \text{min_max}(2,j)) / (\text{min_max}(3,j) - \text{min_max}(2,j)) \times 100,$$

$$\text{if } \text{min_max}(2,j) < \text{GraphData}(i,j) \leq \text{min_max}(3,j) \quad \dots (18c)$$

$$\text{DiagArray}(i,j) = +100, \text{ if } \text{GraphData}(i,j) > \text{min_max}(3,j) \quad \dots (18d)$$

where $\text{min_max}(1,j)$, $\text{min_max}(2,j)$, and $\text{min_max}(3,j)$ represent the minimum, nominal, and maximum acceptable values for sensor j , respectively. A value of 0 indicates optimal operation; values beyond ± 100 flag fault states. This normalization enables a single graphical interface to represent system health across all 30 monitored parameters of the seawater cooling system simultaneously (Simion et al., 2024).

3.3. Sector-Specific Findings

3.3.1. Aviation

A comprehensive system literature review from Pavlyuk & Alomar (2026) investigated 696 articles from 1984 to 2024. It has observed a sharp rise in the adoption of AI after 2018 driven by deep learning. While LSTM (98 articles) and CNN (108 articles) are the most common types of network utilized and applied for remaining useful life (RUL) estimation and visual inspection, respectively, the survey highlights a consistent adoption gap of 4-5 years from the point new AI techniques become commonly recognized in the broader ML domain to their use in aircraft maintenance. This lag is explained by the highly regulated nature of the aviation industry (FAA, EASA). Within the area of Continuing Airworthiness Management (CAWM), RUL prediction was the most widely researched aspect (158 articles), followed by prognostics (80) and reliability analysis (42). There is also a renewed interest in explainable, logic/knowledge based approaches within the aviation industry as a complement to high accuracy but unexplainable deep learning models. The requirement for auditable maintenance decisions within the industry has driven this change.

On an operational level, a real-world AWS native PdM architecture at Southwest Airlines has been described by Patibandla (2024), which uses Amazon Kinesis Data Streams for ingesting real-time data, SageMaker for training and deploying models, SNS for dispatching real-time alerts and QuickSight for visualizations Dashboards. The sensors on an airplane transmit ACARS and QAR data into a maintenance data lake. SageMaker models, trained on this data lake, determine the failure timeline for the components, and allow fleet operators to schedule maintenance prior to failures.

3.3.2. Maritime

Simion et al. (2024) tested their five-module kNN algorithm on the seawater cooling system of an oil tanker-the malfunctioning of this system could lead to a failure of engines. The twelve working states of this system (F1-F12), which ranged from individual components faults (e.g. Sea water chest blockage, pump failure) to multiple components failure (e.g. Multiple pump failure and cooler clogging) were represented as fault array. LINK2PMS connects the artificial intelligence diagnosis to operational maintenance system, showing actions for repair derived from FTA and FMECA within the ship's Planned Maintenance System. A new way to use the sensor values consists on calculation of 10 DELTA parameters, which represent the differences between inlet and outlet of temperature and pressure across a specific component and provide a richer diagnostic signature than raw sensor measurements. Thus, the algorithm is able to distinguish among several fault conditions with identical parameter values.

3.3.3. Automotive

The two automotive related studies also compliment each other. Johari et al. (2026) researches component-level diagnostics: As evidenced by MOMENT, 11 inputs of OBD-II (such as: mass air flow, intake pressure, short-term fuel trim, engine load) are able to be classified into classes of engine air filter health using kNN, achieving a F1 score of 0.983, outperforming SVM (0.925) and Random Forest (0.979), using a dataset of 28,226 samples from a 2014 Suzuki Grand Vitara. Looking at the

correlation matrix, the most highly weighted features were found to be mass air flow and intake pressure; this is intuitively correct given that an engine air filter can cause reduced airflow and disturbed air-fuel ratios if clogged.

Andersson et al. (2025) discusses system-level predictive maintenance within an automotive production plant and implements 3 disparate but complimentary machine learning approaches; 48 hr-ahead failure prediction using LSTM, 2-class (failure/no failure) prediction using SVM and Random Forest (92% accurate), and an autoencoder based unsupervised analysis for anomalies (90% accurate). The result was a 30% reduction in unplanned downtime and a 25% total cost of maintenance, as well as an improvement of machine uptime by 28 percentage points over planned maintenance baselines.

3.4. Common Challenges and Future Directions

Regardless of the domain three constant issues persist. Data Quality, "AI's performance depends on the data from the sensors that train it" (Andersson et al., 2025). Noise, missing or slow measurements reduce the performance of the prediction. Model Interpretability, "deep learning systems are widely recognized as "black boxes", particularly those using LSTMs and auto-encoders" (Pavlyuk & Alomar, 2026). Regulators in transportation can be hesitant to use AI systems for safety-critical issues without this information (Andersson et al., 2025). Regulatory Compliance, regulations require stringent auditing and reporting, something current AI does not perform. Workforce development, "A significant organizational effort must occur in order to reskill and retrain maintenance personnel to use, trust, and react to recommendations generated by an AI system" (Pavlyuk & Alomar, 2026).

Future research should be oriented around: hybrid approaches combining deep learning for predictive and logic/knowledge based systems for explanation capability, federated learning approaches which train models across fleets without centralizing operational sensitive data, and standard benchmark datasets that permit a strong comparison across domains of PdM algorithms. Lack of multi-vehicle model generalization in MOMENT (Johari et al., 2026) and lack of comparison of different installation types for the maritime algorithm (Simion et al., 2024) are also issues to be addressed.

4. Conclusion

In this paper we have reviewed artificial intelligence as applied to PdM in aviation, maritime transport, and automotive manufacturing in light of five peer-reviewed articles published between 2024 and 2026. Each article demonstrated unequivocally that AI based PdM outperform time-based or reactive maintenance in all three sectors.

The math behind the PdM systems are given in equations-Eq.1 FMECA Risk Priority Number to prioritize faults, Eq. 2-4 kNN classifier to detect faults, Eq. 5-9 confusion matrix equations to determine model accuracy, Eq. 10-15 LSTM cells to model time-series RUL, Eq. 16-17 linear regression for degradation prediction, and Eq. 18a-d for normalized health of sensors to reduce bias-and these together present an array of tools to address any number of transport related PdM problem.

These results are indeed noteworthy, as shown by the classification accuracy of >90% for automotive and aviation and 0.983 F1 for diagnosis in air filters with kNN. For prediction of failure 48h ahead with LSTM the accuracy prediction was MAE 0.12%, and for PdM integrated systems a 30% decrease in unexpected stops and 25% cost reduction was observed. The 40-year systematic review of AI literature on aviation from 2016 to 2026 by Pavlyuk & Alomar (2026) supports the conclusion that from 2018 deep learning techniques-particularly LSTMs and CNNs-have superseded classical ML in transportation maintenance as the mainstream approach. Nevertheless, the importance of classical ML should not be undermined as it is useful in supplementing classification tasks which are required to have real-time outputs, or to have explainability, for example.

So, where do we go from here, and how do we bridge the gap between theoretical algorithmic advantage and the necessary operational implementation? Firstly, our models need to be made more explainable to meet regulatory requirements. Secondly, a focus should be placed on developing sensors, and data pipelines that can provide a substantial stream of credible data. Third, structured training programs are necessary so that maintenance operators are enabled and skilled to analyze the information delivered by the AI. Autonomous vessels, aircraft of a new generation and complex automated factories are no longer concepts for the future, but the present. For these vehicles and systems to be safe, efficient and ecologically friendly, AI based predictive maintenance is bound to become obligatory rather than optional.

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